Band-1 Lonely Runner: n=9 Tight Family + Denser Exhaustive Extension

Collapse H-1.05 follow-on — finite exact-Fraction checks only (NOT full LRC)

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2026-09-14

Document class. Scientific computational lab note (Band-1; collapse **H-1.05**).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parent. `2026-09-11` Lonely Runner finite-n lab (tight $k \leq 7$; exhaustive through $n = 8$; 0 failures).
Lab. `/workspace/mission-control/experiments/band1/lonely_runner_n9_extension/`
Cutoff. 2026-09-14.

Honesty stamp (non-negotiable).
This memo is a **finite computational extension** only. It **does not** prove the Lonely Runner Conjecture. It **does not** stamp Clay-closed. It **does not** claim a finished Theory of Everything. GATE-04 collection remains **LOCKED**. Classical computer-assisted literature for larger n is **not** re-derived here; this note only extends the house exact-Fraction ledger.

Abstract

We extend the 2026-09-11 Lonely Runner finite ledger by (i) certifying the tight family $\{1, \dots, k\}$ through $k = 8$ ($n = 9$ runners) with exact Fraction intersection arithmetic; (ii) densifying exhaustive k-subset scans of $\{1, \dots, M\}$ far past the prior (k, M) table, including a new cell $(k, M) = (8, 14)$; (iii) reporting a primitive (gcd = 1) failure count of **0** on those denser cells; (iv) checking additional sporadic speed tuples for $n = 8, 9$ and two $n = 10$ probes. Across **93{,}056** denser exhaustive tuples and all listed sporadics: **0 failures**. The general Lonely Runner Conjecture remains **OPEN**.

Keywords. Lonely runner; Wills–Cusick; exact Fraction arithmetic; finite exhaustive subsets; Band-1; non-claim.

1. Statement lock

Wikipedia / Wills–Cusick convention. n runners on the unit circle; loneliness distance $1/n$.

Reduced Diophantine form. Fix the target runner at 0. Let $k = n - 1$ positive integer speeds $v_1, \dots, v_k$. The reduced loneliness predicate is

$$\exists\, t \in \mathbb{R} \quad \text{such that} \quad \|v_i t\| \geq \delta := \frac{1}{k+1} = \frac{1}{n} \quad \text{for all } i,$$

where $\|x\| := \operatorname{dist}(x, \mathbb{Z})$.

Method (unchanged from parent). For each v, build the closed set $\{t \in [0, 1] : \|vt\| \geq \delta\}$ as a union of rational intervals; intersect over i with exact `fractions.Fraction` arithmetic. Nonempty intersection $\Leftrightarrow$ predicate holds. Grid max-loneliness is diagnostic only.

Finite lemma (extension). For every speed tuple in the tight table $k = 1..8$, the denser exhaustive (k, M) tables below, and the sporadic extension list, the intersection is nonempty. **One finite lemma only — not LRC.**

2. What changed vs 2026-09-11

Axis	Prior (2026-09-11)	This extension
Tight $\{1..k\}$	$k = 1..7$ ($n \leq 8$)	$k = 1..8$ ($n \leq 9$)
Exhaustive cells	$(1, 40), (2, 30), (3, 20), (4, 15), (5, 13), (6, 12), (7, 11)$	$(8, 14)$ **new**
Exhaustive tuples	5{,}521	**93{,}056**
Failures	0	**0** (incl. primitive gcd = 1)
Sporadics	7 cases through $n = 8$	$+n = 9$ sparse + $n = 10$ probes

3. Unit tests

Smoke re-check of tight $k = 1, 2, 3$: classical witness $t = 1/(k+1)$ attains ML $= \delta$; intersection nonempty. **All PASS.**

4. Tight families $\{1, \ldots, k\}$

k	n	Speeds	δ	Exists?	ML at $t = \delta$	Sample t	Witness comps
1	2	{1}	1/2	PASS	1/2	1/2	1
2	3	{1, 2}	1/3	PASS	1/3	1/3	2
3	4	{1..3}	1/4	PASS	1/4	1/4	2
4	5	{1..4}	1/5	PASS	1/5	1/5	2
5	6	{1..5}	1/6	PASS	1/6	1/6	4
6	7	{1..6}	1/7	PASS	1/7	1/7	2
7	8	{1..7}	1/8	PASS	1/8	1/8	4
8	**9**	{1..8}	1/9	**PASS**	1/9	1/9	**6**

New certified tight case. $k = 8$, $n = 9$: classical $t = 1/9$ saturates δ; intersection has 6 closed components.

5. Denser exhaustive k-subsets of $\{1, \ldots, M\}$

All k-subsets of $\{1, \ldots, M\}$ checked with exact intersection. Primitive $= \gcd(v_1, \ldots, v_k) = 1$.

k	n	Prior M	New M	Tuples	Pass	Fail	Prim. fail	Wall (s)
1	2	40	**200**	200	200	0	0	0.02
2	3	30	**80**	3{,}160	3{,}160	0	0	5.14
3	4	20	**40**	9{,}880	9{,}880	0	0	8.28
4	5	15	**28**	20{,}475	20{,}475	0	0	14.36
5	6	13	**22**	26{,}334	26{,}334	0	0	17.74
6	7	12	**18**	18{,}564	18{,}564	0	0	12.18
7	8	11	**16**	11{,}440	11{,}440	0	0	7.74
8	**9**	—	**14**	**3{,}003**	**3{,}003**	**0**	**0**	2.08
Σ				**93{,}056**	**93{,}056**	**0**	**0**	≈ 67

Headline. Denser exhaustive failure count **0**; primitive failure count **0**; first house exhaustive cell at $n = 9$ ($k = 8$, $M = 14$) **PASS**.

6. Sporadic / probe extension

Case	Speeds	n	Exists?	Sample ML
$n = 8$ recheck	$\{1, 4, 5, 6, 7, 11, 13\}$	8	PASS	1/8
$n = 8$ recheck	$\{1, 2, 3, 4, 5, 7, 12\}$	8	PASS	1/8
$n = 9$ sparse	$\{1, 2, 3, 4, 5, 6, 8\}$	9	PASS	9/64
$n = 9$ sparse	$\{1, 2, 3, 4, 5, 7, 11\}$	9	PASS	1/8
$n = 9$ sparse	$\{1, 2, 3, 4, 6, 8, 12\}$	9	PASS	9/64
$n = 9$ sparse	$\{1, 3, 4, 5, 7, 8, 11\}$	9	PASS	7/44
$n = 9$ sparse	$\{1, 2, 4, 5, 7, 10, 13\}$	9	PASS	1/8
$n = 9$ sparse	$\{1, 4, 5, 6, 7, 8, 15\}$	9	PASS	17/112
$n = 9$ sparse	$\{2, 3, 4, 5, 6, 7, 9\}$	9	PASS	23/144
$n = 9$ Fib-ish	$\{1, 2, 3, 5, 8, 13, 21\}$	9	PASS	2/13
$n = 10$ tight	$\{1..9\}$	10	PASS	1/10
$n = 10$ sparse	$\{1, 2, 3, 4, 5, 6, 7, 11\}$	10	PASS	5/42

All **12** sporadic/probe cases **PASS**. The $n = 10$ rows are single-tuple probes, not exhaustive cells.

7. Method limits (explicit)

1. Only concrete finite integer speed tuples are checked.
2. Exhaustive scan is over k-subsets of $\{1..M\}$ for the listed (k, M) only — not all integer k-tuples of unbounded size.
3. Real speeds are not scanned (classical integer reduction is assumed, not re-proved).
4. This does **not** reproduce Rosenfeld / Trakulthongjai-style computer-assisted proofs for large n.
5. Grid max-loneliness is floating; existence uses exact intersection only.

8. Literature status (ledger only)

The Lonely Runner Conjecture remains **OPEN** in general. Classical small-n theorems and modern computer-assisted extensions exist in the literature; this house note does **not** claim to advance that published frontier. It advances the **house exact-Fraction finite ledger** only.

9. Artifacts

Path	Role
`experiments/band1/lonely_runner_n9_extension.py`	Extension driver
`experiments/band1/lonely_runner_n9_extension_core.py`	Exact intersection core (from parent)

Path	Role
`experiments/band1/lonely_runner_n9_extension/results.json`	Machine-readable certificate
Parent `labs/lonely_runner/`	2026-09-11 baseline

10. Non-claims

- Lonely Runner Conjecture **not** solved.
- No Clay stamp; no Millennium language.
- No finished ToE; **GATE-04 LOCKED**.
- Non-finding outside listed (k, M) cells is **not** a counterexample search.
- $n = 10$ probes are **not** an exhaustive $n = 10$ ledger.

11. One-line summary

Extension PASS: tight $k = 1..8$; denser exhaustive **93{,}056** / **0** fail (primitive **0**); sporadic extension PASS. LRC remains **OPEN**. GATE-04 **LOCKED**.